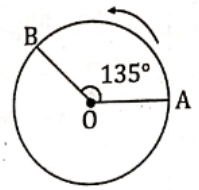
Question	A body moves 6m north, 8m east and 10m vertically upwards, what is its resultant displacement from its initial position?
Type	multiple choice
Option	$10\sqrt{2}\text{m}$
Option	10m
Option	$\frac{10}{\sqrt{2}}\text{m}$
Option	$10\times 2\text{m}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A uniform velocity of $45\text{km}/\text{h}$. The time taken by the train to cross a bridge of length 850 meters is:
Type	multiple choice
Option	58sec
Option	68sec
Option	80sec
Option	92sec
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A person moves 30m north and then $20\sqrt{2}\text{m}$ towards east and finally $30\sqrt{2}\text{m}$ in south-west direction. The displacement of the person from the origin will be:
Type	multiple choice
Option	10m along north
Option	10m along south
Option	10m along west
Option	Zero
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A wheel of radius 1 meter rolls forward half a revolution on a horizontal ground. The magnitude of the displacement of the point of the wheel initially in contact with the ground is:
Type	multiple choice
Option	$2\pi\text{m}$
Option	$\sqrt{2}\pi\text{m}$
Option	$\sqrt{\pi^2+4}\text{m}$
Option	πm
Answer	3
Solution	
Positive Marks	4
Negative Marks	1

Question	The three initial and final position of a man on the x-axis are given as: (i) (2m to 7m), (ii) (3m to -3m), and (iii) (-7m to 3m). Which pair gives the negative displacement?
Type	multiple choice
Option	(i) and (ii)
Option	(i) and (iii)
Option	(ii) and (iii)
Option	(i) and (ii)
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A particle is moving along a circle such that it completes one revolution in 40 seconds. In 2 minutes 20 seconds, the ratio $\frac{\lvert\text{displacement}\rvert}{\text{distance}}$ is:
Type	multiple choice
Option	0
Option	$\frac{1}{7}$
Option	$\frac{2}{7}$
Option	$\frac{1}{11}$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A person moved from A to B on a circular path as shown in figure. If the distance travelled by him is 40m, then the magnitude of displacement would be: (Given $\cos 135^\circ = 0.7$) 
Type	multiple choice
Option	42m
Option	47m
Option	19m
Option	40m
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The numerical ratio of displacement to distance for a moving object is:
Type	multiple choice
Option	Always less than 1
Option	Always equal to 1
Option	Always more than 1
Option	Equal to or less than 1

Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	If a car covers 2/5 of the total distance with v_1 speed and $3/5$ distance with v_2 Then average speed is:
Type	multiple choice
Option	$\frac{1}{2}\sqrt{v_1 v_2}$
Option	$\frac{v_1+v_2}{2}$
Option	$\frac{2v_1v_2}{v_1+v_2}$
Option	$\frac{5v_1v_2}{3v_1+2v_2}$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A bus driver is driving along a side-road at 45m/s when suddenly he encounters a puppy run across the street. Thus, he rapidly slows down to 3m/s within a time span of 1.5sec. Calculate the average acceleration of the bus.
Type	multiple choice
Option	28m/s^2
Option	14m/s^2
Option	-14m/s^2
Option	-28m/s^2
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	At any instant, the velocity and acceleration of a particle moving along a straight line are v and a respectively. The speed of the particle is increasing if:
Type	multiple choice
Option	$v>0, a>0$
Option	$v<0, a>0$
Option	$v>0, a<0$
Option	$v>0, a\le 0$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The position of a particle varies with time according to the relation $x=3t^2+5t^3+7t$, where x is in m and t is in s. Find average velocity during time interval 0 to 5s.
Type	multiple choice
Option	37m/s

Option	147m/s
Option	123m/s
Option	153m/s
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The position of an object moving along x-axis is given by $x = a + bt^2$, where $a = 8.5\text{m}$ and $b = 2.5\text{m/s}^2$ and t is measured in seconds. The average velocity of the object between $t = 2\text{s}$ and $t = 4\text{s}$ is:
Type	multiple choice
Option	5ms^{-1}
Option	10ms^{-1}
Option	15ms^{-1}
Option	20ms^{-1}
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A cyclist moving on a circular track of radius 40m completes half a revolution in 40sec. Its average velocity is:
Type	multiple choice
Option	Zero
Option	$4\pi\text{m/s}$
Option	2m/s
Option	$6\pi\text{m/s}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A man walks on a straight road from his home to a market 2.5km away with a speed of 5km/h. Finding the market closed, he instantly turns and walks back home with a speed of 7.5km/h. The average speed of the man over the interval time of 0 to 40min is equal to:
Type	multiple choice
Option	5km/h
Option	$\frac{25}{4}\text{km/h}$
Option	$\frac{30}{4}\text{km/h}$
Option	$\frac{45}{8}\text{km/h}$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1

Question	The displacement is given by $x = 2t^2 + t + 5$. The acceleration at $t = 2s$ is
Type	multiple choice
Option	$4\ m/s^2$
Option	$8\ m/s^2$
Option	$10\ m/s^2$
Option	$5\ m/s^2$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The velocity of a body depends on time according to the equation $v = 20 + 0.1t^2$. The body is undergoing
Type	multiple choice
Option	Uniform acceleration
Option	Uniform retardation
Option	Non-uniform acceleration
Option	Zero acceleration
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The displacement of a particle is given by $y = a + bt + ct^2 - dt^4$. The initial velocity and acceleration are respectively
Type	multiple choice
Option	b, -4d
Option	-b, 2c
Option	b, 2c
Option	2c, -4d
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The relation between time and distance is $t = \alpha x^2 + \beta x$, where α and β are constants. The retardation is (v is velocity)
Type	multiple choice
Option	$2\alpha v^3 2\beta v^3$
Option	$2\alpha v^2$
Option	$2\beta v^3$
Option	1
Answer	
Solution	
Positive Marks	4
Negative Marks	1
Question	The acceleration 'a' in m/s^2 of a particle is given by $a = 3t^2 + 2t + 2$ where t is the time. If the particle starts out with a velocity $u = 2m/s$ at $t = 0$, then the velocity at the end of 2 second is
Type	multiple choice
Option	12 m/s
Option	18 m/s
Option	27 m/s
Option	36 m/s
Answer	2
Solution	

Positive Marks	4
Negative Marks	1
Question	The velocity of a body moving along the x-axis varies with x as $v = (x^2 - x^4)$ m/s. Find the acceleration of the body at x = 2 m, if the acceleration is defined as a = v(dx/dx)
Type	multiple choice
Option	132 m/s^2
Option	32 m/s^2
Option	8 m/s^2
Option	4 m/s^2
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A particle moves along the x-axis according to $s = t^3 - (15/2)t^2 + 12t + 5$, where symbols have their usual meaning. At what time, velocity changes its direction?
Type	multiple choice
Option	4 seconds
Option	3 seconds
Option	2 seconds
Option	5 seconds
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Given velocity v(t) = 2t + 3. Assume s(t) is measured in meters and t is measured in seconds. If s(0) = 0, the position s(4) at = 4s is: (Given, v = (ds/dt)
Type	multiple choice
Option	30
Option	31
Option	32
Option	33
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A motor car moving with a uniform speed of 20 m/sec comes to stop on the application of brakes after travelling a distance of 10 m. Its acceleration is
Type	multiple choice
Option	20 m/sec^2
Option	-20 m/sec^2
Option	-40 m/sec^2
Option	+2 m/sec^2
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The velocity of a body moving with a uniform acceleration of 2m/sec^2 is 10 m/sec. Its velocity after an interval of 4sec is
Type	multiple choice
Option	12 m/sec
Option	14m/sec
Option	16m/sec
Option	18 m/sec

Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A particle starting from rest moving with constant acceleration travels a distance x in first 2 seconds and a distance y in next two seconds. Then
Type	multiple choice
Option	y = x
Option	y = 2x
Option	y = 3x
Option	y = 4x
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	If a car at rest accelerates uniformly to a speed of 144 km/h in 20s. Then it covers a distance of
Type	multiple choice
Option	20 m
Option	400 m
Option	1440 m
Option	2880 m
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The velocity of a bullet is reduced from 200 m/s to 100 m/s while travelling through a wooden block of thickness 10 cm. The retardation, assuming it to be uniform will be
Type	multiple choice
Option	10×10^4 m/s^2
Option	12×10^4 m/s^2
Option	13.5×10^4 m/s^2
Option	15×10^4 m/s^2
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A particle is moving along x-axis with initial velocity 20 m/s. It experiences a retardation of 20 m/s^2 along x-axis. Distance travelled by particle in 2 s is
Type	multiple choice
Option	20 m
Option	15 m
Option	10 m
Option	5 m
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Speed of two identical cars are u and 4u at a specific instant. The ratio of the respective distances in which the two cars are stopped from that instant is:
Type	multiple choice
Option	1 : 1
Option	1 : 4

Option	1 : 8
Option	1 : 16
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A ball is thrown upward with a velocity of $100m/s$. It will reach the ground after (Take $g = 10m/s^2$): -
Type	multiple choice
Option	$10s$
Option	$20s$
Option	$5s$
Option	$40s$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	If a ball is thrown vertically upwards with velocity of $40m/s$. Its velocity after two seconds will be (Take $g = 10m/s^2$): -
Type	multiple choice
Option	$10m/s$
Option	$20m/s$
Option	$30m/s$
Option	$40m/s$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A body projected vertically upwards with a velocity u returns to the starting point in 4 seconds. If $g = 10m/sec$, the value of u is
Type	multiple choice
Option	$5m/sec$
Option	$10m/sec$
Option	$15m/sec$
Option	$20m/sec$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A particle is thrown vertically upward. Its velocity at half of the maximum height is $10m/s$. The maximum height attained by it is ($g = 10m/s^{-2}$): -
Type	multiple choice
Option	$8m$
Option	$20m$
Option	$10m$
Option	$16m$
Answer	3

Solution	
Positive Marks	4
Negative Marks	1
Question	A particle projected vertically upwards with a velocity u from ground. When it returns to the ground its:
Type	multiple choice
Option	average speed is $u/2$
Option	Average speed is u
Option	Average speed is zero
Option	None of these
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	When a ball is thrown vertically up with velocity v_0, it reaches a maximum height 'h'. If one wishes to triple the maximum height then the ball should be thrown with velocity-
Type	multiple choice
Option	$\sqrt{3}v_0$
Option	$3v_0$
Option	$9v_0$
Option	$3/2v_0$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A person throws ball into air vertically upwards at regular intervals of time of one second. The next ball is thrown when the velocity of the ball thrown earlier becomes zero. The height to which the balls rise is: (Assume $g = 10m/s^{-2}$)
Type	multiple choice
Option	$5m$
Option	$10m$
Option	$7.5m$
Option	$20m$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A particle is thrown up vertically with a speed 'v_1' in air. It takes time t_1 in upward journey and $t_2 (> t_1)$ in the downward journey and returns to the starting point with a speed v_2. Then
Type	multiple choice
Option	$v_1 = v_2$
Option	$v_1 < v_2$
Option	$v_1 > v_2$
Option	Data is insufficient
Answer	3
Solution	
Positive Marks	4

Negative Marks	1
Question	With what speed should a body be thrown upward so that the distance traversed in 5th second and 6th second are equal?
Type	multiple choice
Option	$58.4m/s$
Option	$49m/s$
Option	$\sqrt{98}m/s$
Option	$98m/s$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A rocket is fired upward from the earth's surface such that it creates an acceleration of $19.6m/s^2$. If after 5 sec its engine is switched off, the maximum height of the rocket from earth's surface would be
Type	multiple choice
Option	$245m$
Option	$490m$
Option	$980m$
Option	$735m$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A body dropped from a tower reaches the ground in 4s. The height of the tower is about (Take $g = 10m/s^2$):
Type	multiple choice
Option	$80m$
Option	$20m$
Option	$160m$
Option	$40m$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A body is released from the top of a tower of height H metres. It takes t time to reach the ground. Where is the body $\frac{t}{2}$ time after the release (Take $g = 10m/s^2$):
Type	multiple choice
Option	At $\frac{H}{2}$ metres from ground
Option	At $\frac{H}{4}$ metres from ground
Option	At $\frac{3H}{4}$ metres from the ground

Option	At $\frac{H}{6}$ metres from the ground
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A body falling from height 'h' takes t_1 time to reach the ground. The time taken to cover the first half of the height is (Take $g = 10m/s^2$)
Type	multiple choice
Option	$t_2 = \frac{t_1}{\sqrt{2}}$
Option	$t_1 = \frac{t_2}{\sqrt{2}}$
Option	$t_2 = \sqrt{3}t_1$
Option	None of these
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Two bodies of different masses m_a and m_b are dropped from two different heights a and b. The ratio of the time taken by the two to cover these distances are
Type	multiple choice
Option	$a:b$
Option	$b:a$
Option	$\sqrt{a}:\sqrt{b}$
Option	$a^2:b^2$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A ball is dropped downwards. After 1 second another ball is dropped downwards from the same point. What is the distance between them after 3 seconds (Take $g = 10m/s^2$).
Type	multiple choice
Option	$25m$
Option	$29m$
Option	$50m$
Option	$9.8m$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1